

6.3 ORTHOGONAL PROJECTIONS

The orthogonal projection of a point in $\mathbb{R}^2$ onto a line through the origin has an important analogue in $\mathbb{R}^n$. Given a vector y and a subspace W in $\mathbb{R}^n$, there is a vector $\bar{y}$ in W such that (1) $\bar{y}$ is the unique vector in W for which $y - \bar{y}$ is orthogonal to W , and (2) $\bar{y}$ is the unique vector in W closest to y . See Figure 1. These two properties of $\bar{y}$ provide the key to finding least-squares solutions of linear systems, mentioned in the introductory example for this chapter. The full story will be told in Section 6.5.

To prepare for the first theorem, observe that whenever a vector y is written as a linear combination of vectors $u_1, \dots, u_n$ in $\mathbb{R}^n$, the terms in the sum for y can be grouped into two parts so that y can be written as

$$y = z_1 + z_2$$

where z_1 is a linear combination of some of the u_i and z_2 is a linear combination of the rest of the u_i . This idea is particularly useful when $\{u_1, \dots, u_n\}$ is an orthogonal basis. Recall from Section 6.1 that $W^\perp$ denotes the set of all vectors orthogonal to a subspace W .

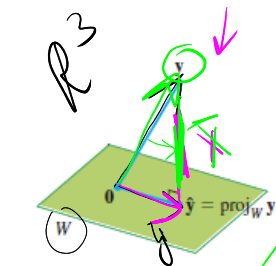
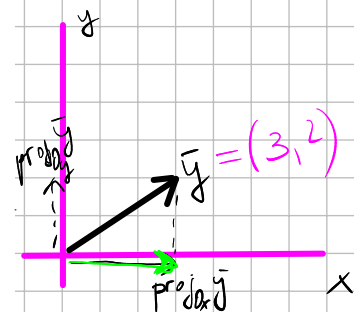


FIGURE 1



$$\hat{y} + \bar{x} = y$$

$$\bar{x} = y - \bar{y}$$

$$\bar{y} = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$$

$$\bar{y} = \bar{z}_1 + \bar{z}_2$$

$$z_1 = c_1 u_1 + c_2 u_2$$

$$z_2 = c_3 u_3 + \dots + c_n u_n$$

Orthogonal basis $\mathbb{R}^2$

$$\begin{bmatrix} i \\ 0 \end{bmatrix} \cdot \begin{bmatrix} j \\ 1 \end{bmatrix}$$

$$\bar{y} = 3 \cdot i + 2 \cdot j$$

$$\text{proj}_{\partial_x} \bar{y} = 3i$$

$$\text{proj}_{\partial_y} \bar{y} = 2j$$

EXAMPLE 1 Let $\{u_1, \dots, u_5\}$ be an orthogonal basis for $\mathbb{R}^5$ and let

$$y = c_1 u_1 + \dots + c_5 u_5$$

$$y = (c_1, c_2, \dots, c_5)$$

Consider the subspace $W = \text{Span}\{u_1, u_2\}$, and write y as the sum of a vector z_1 in W and a vector z_2 in $W^\perp$.

$$y = z_1 + z_2$$

$$z_1 = c_1 u_1 + c_2 u_2 \in W$$

$$z_2 = c_3 u_3 + c_4 u_4 + c_5 u_5 \in W^\perp$$

$$z_1 \perp W^\perp, \quad z_2 \perp W \rightarrow \text{span}\{u_1, u_2\}$$

$$\begin{aligned} z_1 &\perp u_3 \\ z_1 &\perp u_4 \\ z_1 &\perp u_5 \end{aligned}$$

$$\begin{aligned} z_2 &\perp u_1 \\ z_2 &\perp u_2 \\ z_1 \cdot u_3 &= (c_1 u_1 + c_2 u_2) \cdot u_3 = \\ &= c_1 (u_1 \cdot u_3) + c_2 (u_2 \cdot u_3) = 0 \end{aligned}$$

The next theorem shows that the decomposition $y = z_1 + z_2$ in Example 1 can be computed without having an orthogonal basis for $\mathbb{R}^n$. It is enough to have an orthogonal basis only for W .

THEOREM 8

The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then each y in $\mathbb{R}^n$ can be written uniquely in the form

$$y = \hat{y} + z \quad (1)$$

where $\hat{y}$ is in W and z is in $W^\perp$. In fact, if $\{u_1, \dots, u_p\}$ is any orthogonal basis of W then

$$\hat{y} = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \dots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p \quad (2)$$

and $z = y - \hat{y}$.

The vector $\hat{y}$ in (1) is called the **orthogonal projection of y onto W** and often is written as $\text{proj}_W y$. See Figure 2. When W is a one-dimensional subspace, the formula for $\hat{y}$ matches the formula given in Section 6.2.

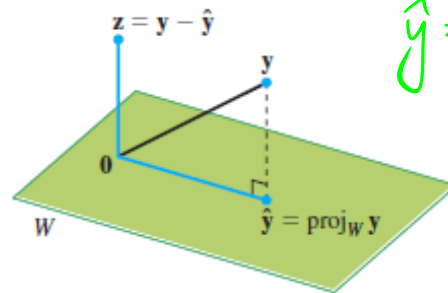


FIGURE 2 The orthogonal projection of y onto W .

$$\hat{y} = \sum_{i=1}^p \frac{y \cdot u_i}{u_i \cdot u_i} u_i$$

The uniqueness of the decomposition (1) shows that the orthogonal projection $\hat{y}$ depends only on W and not on the particular basis used in (2).

EXAMPLE 2 Let $u_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, and $y = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Observe that $\{u_1, u_2\}$ is an orthogonal basis for $W = \text{Span}\{u_1, u_2\}$. Write y as the sum of a vector in W and a vector orthogonal to W .

$$y = z_1 + z_2$$

$$z_1 \in W = \text{span}\{u_1, u_2\}$$

$$z_2 = W^\perp \text{span}\{u_1, u_2\}$$

$$z_1 = \underset{\uparrow}{C_1} \cdot u_1 + \underset{\uparrow}{C_2} \cdot u_2$$

$$C_1 = \frac{y \cdot u_1}{u_1 \cdot u_1}$$

$$C_2 = \frac{y \cdot u_2}{u_2 \cdot u_2}$$

$$u_1^2 = 2^2 + 5^2 + 1^2 = 30,$$

$$u_2^2 = 2^2 + 1^2 + 1^2 = 6$$

$$y \cdot u_1 = 2 \cdot 1 + 5 \cdot 2 - 1 \cdot 3 = 9;$$

$$y \cdot u_2 = -2 \cdot 1 + 1 \cdot 2 + 1 \cdot 3 = 3$$

$$C_1 = \frac{9}{30} = \frac{3}{10}$$

$$C_2 = \frac{3}{6} = \frac{1}{2}$$

$$z_1 = 0.3 \cdot u_1 + 0.5 \cdot u_2 = \frac{3}{10} \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}$$

$$z_2 = y - z_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} =$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

$$y = z_1 + z_2; \quad z_1 = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = 0.3 \cdot u_1 + 0.5 \cdot u_2$$

$$z_2 = \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$